Reflection in n-Dimensional Euclidean Space

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Linear Algebra · Geometry

Abstract. *We develop a systematic treatment of geometric reflection across dimensions. Beginning with 2D reflections over axes and arbitrary lines, we extend to 3D reflections over axes, planes, and points, presenting each case with its transformation matrix. We then derive the general n-dimensional form using the Householder reflection framework. A compact notation* $\mathrm{Ref}^{\mathbb{R}^n}_{reflector}$ *is introduced throughout.*

1. Introduction and Notation

Definition 1 (Reflector)**.** A *reflector* is the geometric object with respect to which a reflection is performed. In $\mathbb{R}^n$, reflectors include axes, hyperplanes, the origin, and points.

We introduce the notation $\mathrm{Ref}^{\mathbb{R}^n}_{\text{reflector}}(V)$ to encode three things simultaneously: the dimension $\mathbb{R}^n$, the reflector, and the object V being reflected.

Since reflection through the origin is consistent across all dimensions, we omit the subscript 0 and write $\mathrm{Ref}(V) := \mathrm{Ref}_0(V)$ for conciseness.

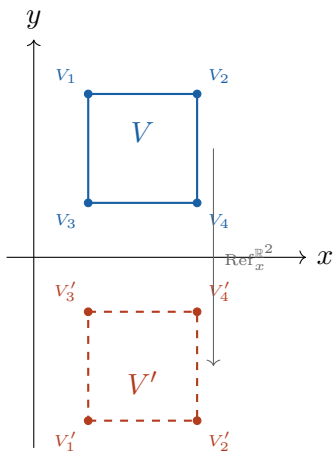
2. Reflection in $\mathbb{R}^2$

In $\mathbb{R}^2$, reflecting an object means flipping it over an arbitrary line. The standard reflectors are the x-axis, the y-axis, and lines of the form $y = x$, $y = -x$, or more generally $y = mx + c$.

2.1 Standard axis reflections

Consider a square V with vertices:

$$V = \{(1,3), (3,3), (1,1), (3,1)\}$$



To reflect over the x-axis ($y = 0$), multiply by the transformation matrix:

$$\mathrm{Ref}_x^{\mathbb{R}^2}(V) = V \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \{(1,-3),(3,-3),(1,-1),(3,-1)\}$$

The remaining standard 2D reflection matrices are:

$$\mathrm{Ref}_y^{\mathbb{R}^2}(V) = V \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad \mathrm{Ref}_{y=x}^{\mathbb{R}^2}(V) = V \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \qquad \mathrm{Ref}_{y=-x}^{\mathbb{R}^2}(V) = V \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$$

Reflection through the origin produces the same result as a 180 rotation:

$$\mathrm{Ref}_0^{\mathbb{R}^2}(V) = V \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

2.2 Reflection over an arbitrary line $y = mx + c$

Reflection through x-axis, y-axis, $y = x$, and $y = -x$ are special cases. For a general line $y = mx + c$, we derive the reflected point in two steps.

Let $P(a, b)$ be the point to reflect and let (x_1, y_1) be the closest point on the line to P (the foot of the perpendicular).

Step 1: Find the perpendicular foot. The perpendicular from P to the line has slope $-1/m$, giving:

$$y_1 - b = -\tfrac{1}{m}(x_1 - a) \quad \text{and} \quad y_1 = mx_1 + c$$

Solving simultaneously:

$$x_1 = \frac{a + mb - mc}{m^2 + 1}, \qquad y_1 = m \cdot \frac{a + mb - mc}{m^2 + 1} + c$$

Step 2: Find the reflected point. Let $P'(a', b')$ be the reflection of P. Since (x_1, y_1) is the midpoint of P and P':

$$a' = 2x_1 - a = \frac{(1 - m^2)a + 2mb - 2mc}{m^2 + 1}, \qquad b' = 2y_1 - b = \frac{(m^2 - 1)b + 2ma + 2c}{m^2 + 1}$$

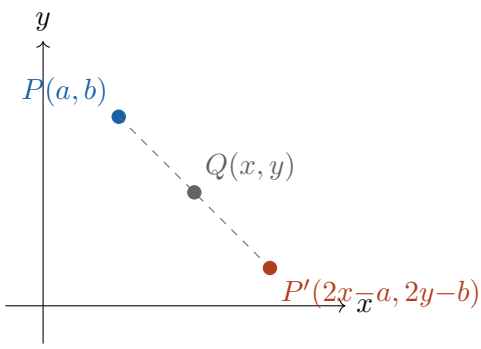
Therefore:

$$P' = \mathrm{Ref}_Q^{\mathbb{R}^2}(P) = \begin{pmatrix} \dfrac{(1-m^2)a + 2m(b-c)}{m^2+1} \\ \dfrac{(m^2-1)b + 2ma + 2c}{m^2+1} \end{pmatrix} \quad \text{where } y = mx + c$$

2.3 Reflection through an arbitrary point

A point $P(a, b)$ reflected through a centre point $Q(x, y)$ satisfies $(P + P')/2 = Q$, giving:

$$P' = 2Q - P = \mathrm{Ref}_Q^{\mathbb{R}^2}(P) = \begin{pmatrix} 2x - a \\ 2y - b \end{pmatrix}$$



2.4 Remark on reflecting through an arbitrary function

Reflecting through x-axis, y-axis, $y = x$, and $y = -x$ are all well-defined because each is a line with a unique perpendicular from any point. Extending reflection to an arbitrary function $f(x)$ is more subtle: a point may have multiple closest points on the curve, making the reflected image ambiguous. In such cases, the definition of reflection depends on additional choices made by the person performing the calculation.

3. Reflection in $\mathbb{R}^3$

In $\mathbb{R}^3$, reflection can occur over an axis, a plane, or the origin. The table below summarises all standard cases.

Reflector	**Transformation**	**Matrix**
Origin	$(x, y, z) \to (-x, -y, -z)$	$\mathrm{Ref}^3 = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$
x-axis	$(x, y, z) \to (x, -y, -z)$	$\mathrm{Ref}^3_x = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$
y-axis	$(x, y, z) \to (-x, y, -z)$	$\mathrm{Ref}^3_y = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$
z-axis	$(x, y, z) \to (-x, -y, z)$	$\mathrm{Ref}^3_z = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$
xy-plane	$(x, y, z) \to (x, y, -z)$	$\mathrm{Ref}^3_{xy} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$
xz-plane	$(x, y, z) \to (x, -y, z)$	$\mathrm{Ref}^3_{xz} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$
yz-plane	$(x, y, z) \to (-x, y, z)$	$\mathrm{Ref}^3_{yz} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$
Point $P(a, b, c)$	$(x, y, z) \to (-x+2a, -y+2b, -z+2c)$	$\mathrm{Ref}^3_P = \begin{pmatrix} -x+2a \\ -y+2b \\ -z+2c \end{pmatrix}$

4. Reflection in $\mathbb{R}^4$

In $\mathbb{R}^4$ there are four hyperplanes to consider: $x_1x_2x_3$, $x_1x_2x_4$, $x_1x_3x_4$, and $x_2x_3x_4$. As an example:

$$\mathrm{Ref}^4_{\bar{x}_4} = \mathrm{Ref}^4_{x_1x_2x_3} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

5. n-Dimensional Generalisation

In $\mathbb{R}^n$, reflection can be expressed uniformly using the identity matrix I and the outer product $\mathbf{x}_i\mathbf{x}_i^\top$.

Reflector	**Matrix**
Origin	$\mathrm{Ref} = -I$
Axis x_i	$\mathrm{Ref}_{x_i} = -I + 2\,\mathbf{x}_i\mathbf{x}_i^\top$
Hyperplane $\bar{x}_i$	$\mathrm{Ref}_{\bar{x}_i} = I - 2\,\mathbf{x}_i\mathbf{x}_i^\top$
Point P	$\mathrm{Ref}_P(V) = -V + 2P$

Here $\mathbf{x}_i$ is the i-th standard basis vector and $\mathbf{x}_i\mathbf{x}_i^\top$ is its outer product (a rank-1 matrix with a single 1 in the (i,i) position and zeros elsewhere). The hyperplane formula $I - 2\mathbf{x}_i\mathbf{x}_i^\top$ is the classical *Householder reflection* matrix.

Remark 1. The axis and hyperplane formulas are complementary: reflecting over an axis flips all coordinates *except* the axis direction, while reflecting over a hyperplane flips only the coordinate *perpendicular* to it. This is captured neatly by the sign difference: $-I + 2\mathbf{x}_i\mathbf{x}_i^\top$ versus $I - 2\mathbf{x}_i\mathbf{x}_i^\top$.